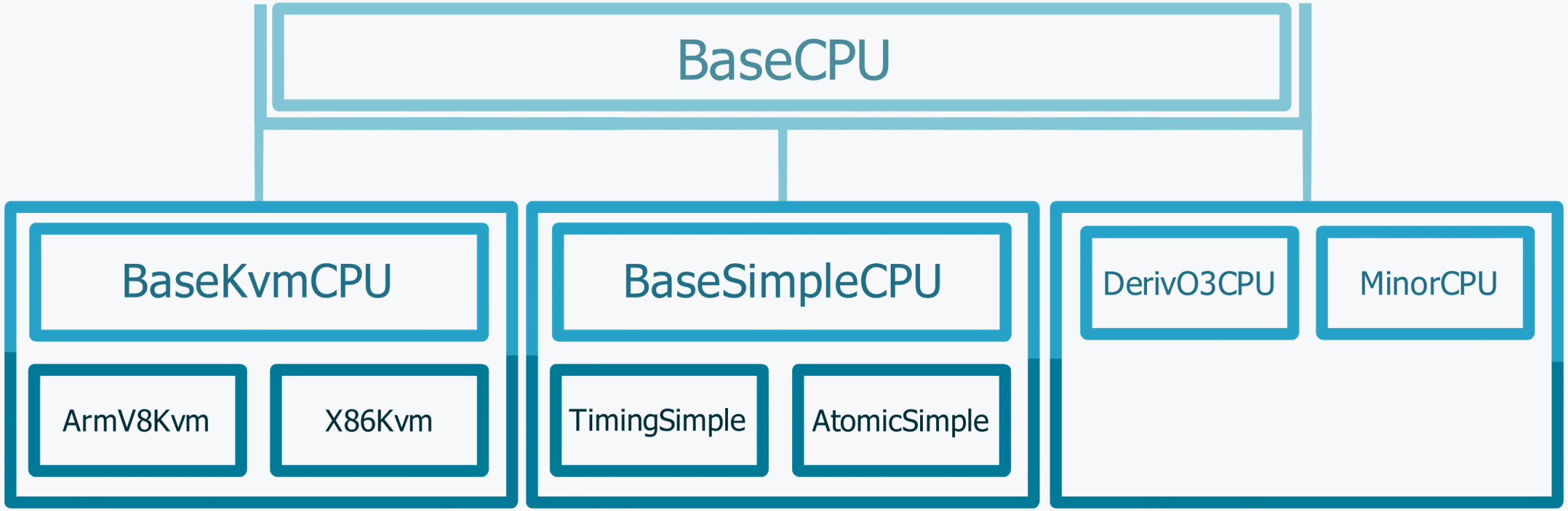
Modeling CPU cores in gem5

Outline

- **Learn about CPU models in gem5**
  - AtomicSimpleCPU, TimingSimpleCPU, O3CPU, MinorCPU, KvmCPU
- Using the CPU models
  - Set up a simple system with two cache sizes and three CPU models
- Look at the gem5 generated statistics
  - To understand the differences between CPU models
- Create a custom processor
  - Change parameters of a processor based on O3CPU

gem5 CPU Models



Simple CPU

SimpleCPU

Atomic

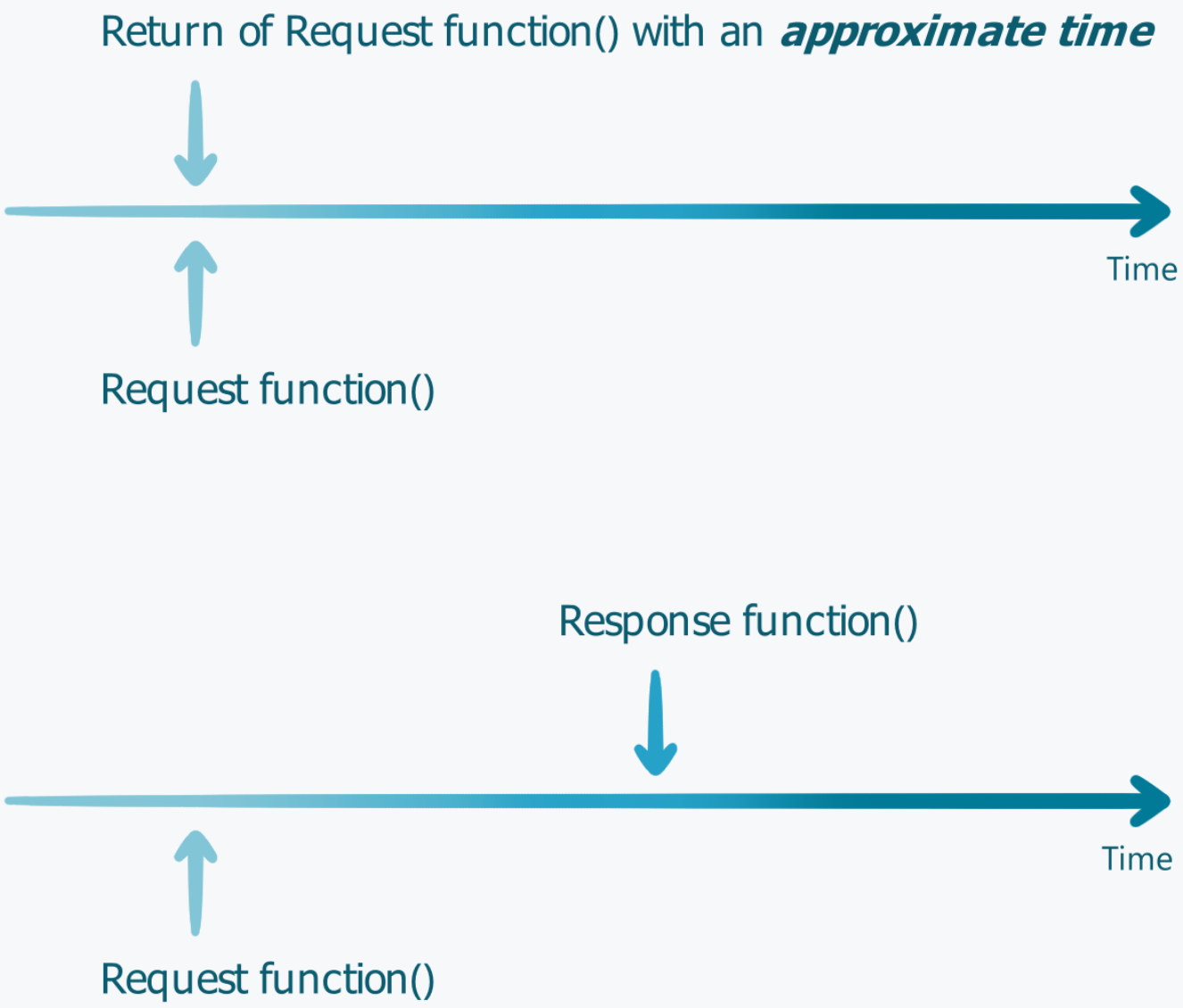
Sequence of nested calls
Use: Warming up, fast-forwarding

Functional

Backdoor access to mem.
(loading binaries)
No effect on coherency states

Timing

Split transactions
Models queuing delay and
resource contention



Other Simple CPUs

AtomicSimpleCPU

- Uses ***Atomic*** memory accesses
  - No resource contentions or queuing delay
  - Mostly used for fast-forwarding and warming of caches

TimingSimpleCPU

- Uses ***Timing*** memory accesses
  - Execute non-memory operations in one cycle
  - Models the timing of memory accesses in detail

O3CPU (Out of Order CPU Model)

- ***Timing*** memory accesses *execute-in-execute* semantics
- Time buffers between stages

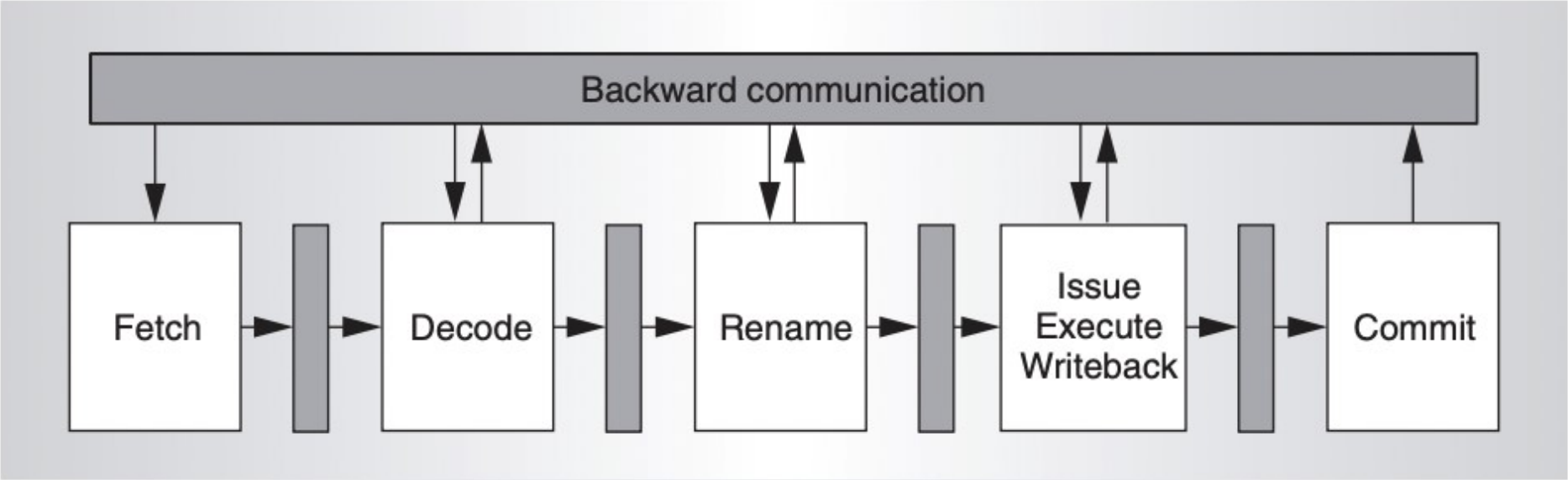


Figure 2. O3CPU pipeline. Shaded boxes represent time buffers.

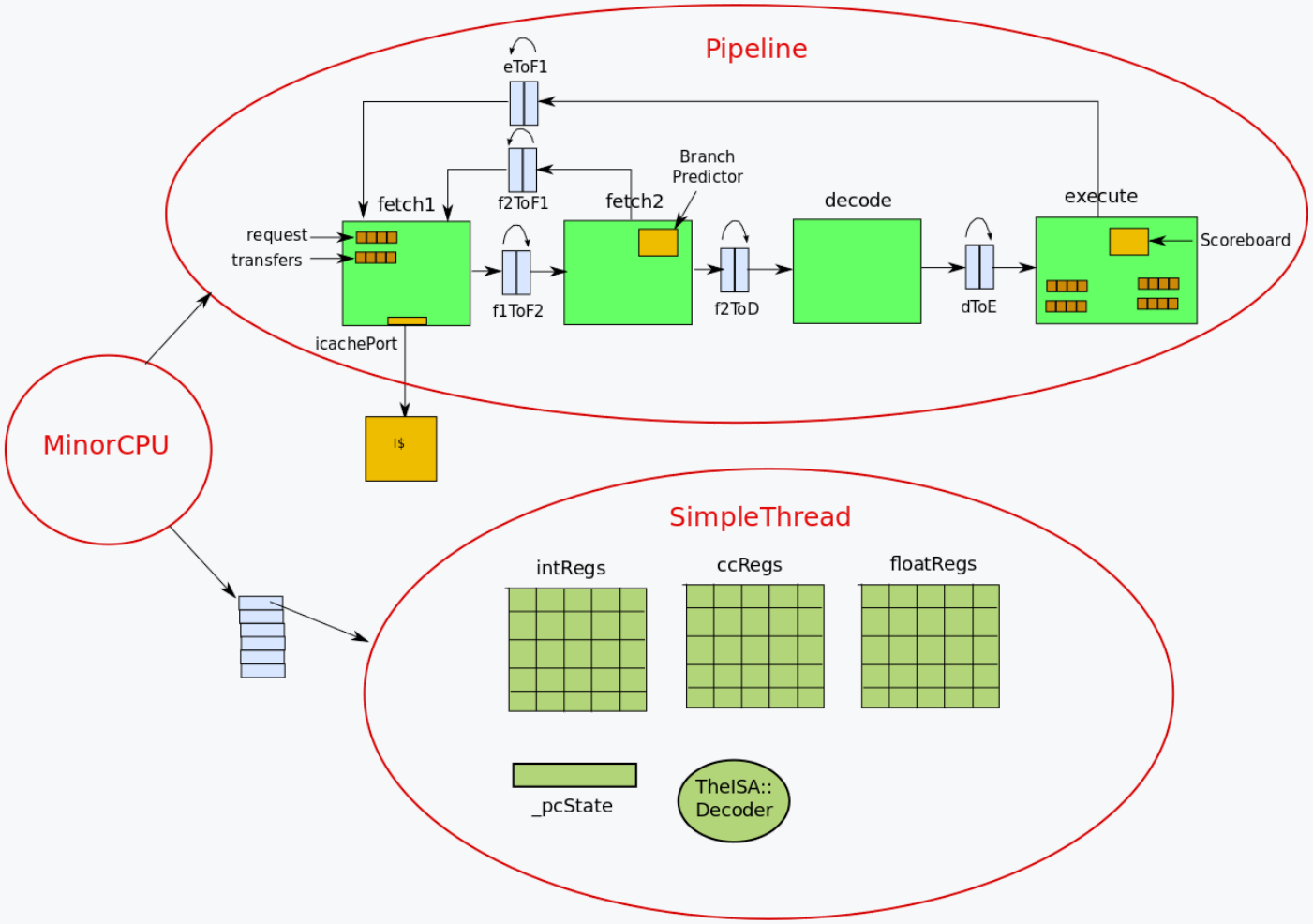
The O3CPU Model has many parameters

src/cpu/o3/BaseO3CPU.py

```
decodeToFetchDelay = Param.Cycles(1, "Decode to fetch delay")
renameToFetchDelay = Param.Cycles(1, "Rename to fetch delay")
...
fetchWidth = Param.Unsigned(8, "Fetch width")
fetchBufferSize = Param.Unsigned(64, "Fetch buffer size in bytes")
fetchQueueSize = Param.Unsigned(
    32, "Fetch queue size in micro-ops per-thread"
)
...
```

Remember, do not update the parameters directly in the file. Instead, create a new *stdlib component* and extend the model with new values for parameters.
We will do this soon.

MinorCPU


Pipeline
eToF1
f2ToF1
Branch
Predictor
fetch1
fetch2
decode
execute
request
transfers
f1ToF2
f2ToD
dToE
Scoreboard
icachePort
I$
MinorCPU
SimpleThread
intRegs
ccRegs
floatRegs
_pcState
TheISA::
Decoder

KvmCPU

- KVM – Kernel-based virtual machine
- Used for native execution on x86 and ARM host platforms
- Guest and the host need to have the same ISA
- Very useful for functional tests and fast-forwarding

Summary of gem5 CPU Models

BaseKvmCPU

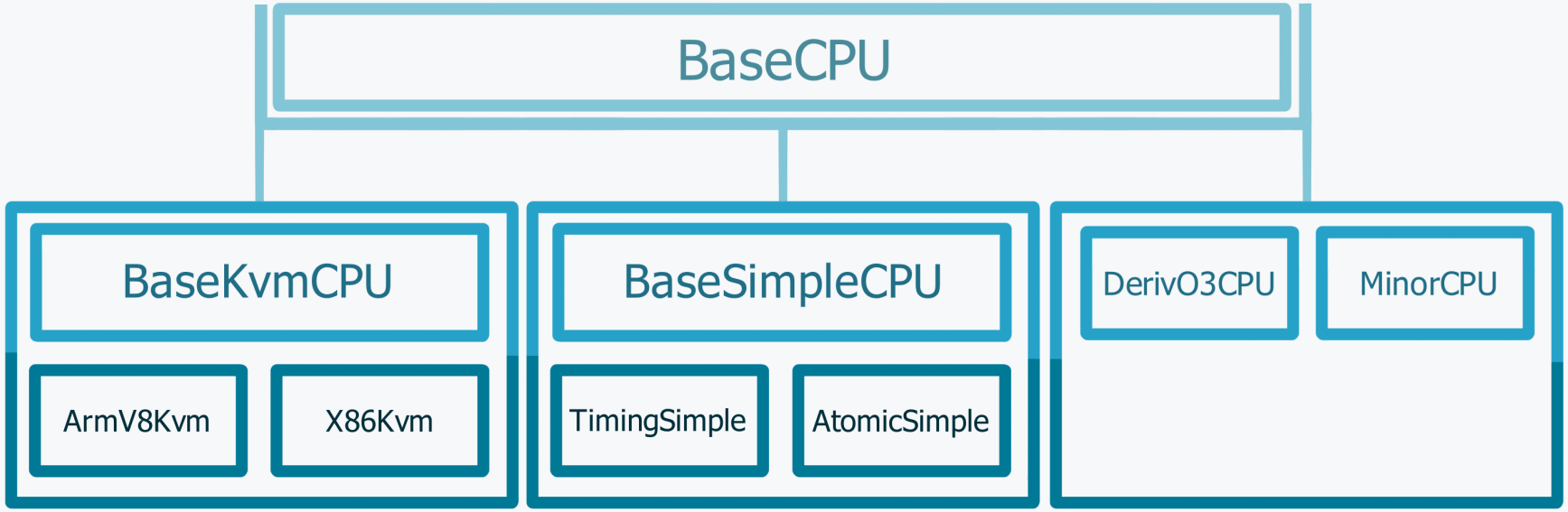
- Very fast
- No timing
- No caches, BP

BaseSimpleCPU

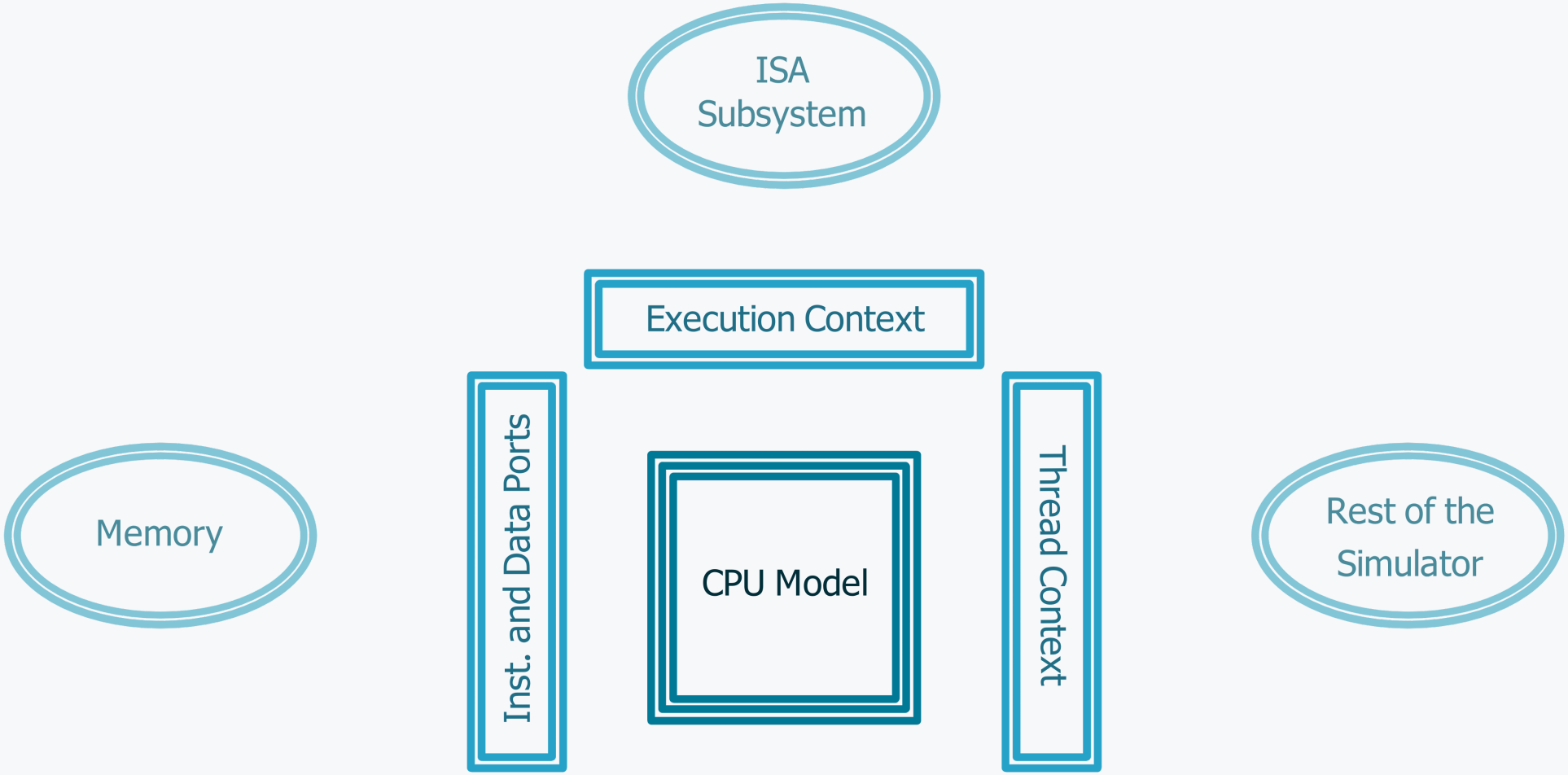
- Fast
- Some timing
- Caches, limited BP

DerivO3CPU and MinorCPU

- Slow
- Timing
- Caches, BP



Interaction of CPU model with other parts of gem5



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Let's use these CPU Models!

Material to use

Start by opening the following file

materials/02-Using-gem5/04-cores/cores.py

Steps

1. Configure a simple system with Atomic CPU
2. Configure the same system with Timing CPU
3. Reduce the cache size
4. Change the CPU type back to Atomic

We will be running a workload called matrix-multiply on **different CPU types and cache sizes**.

Let's configure a simple system with Atomic CPU

materials/02-Using-gem5/04-cores/cores.py

```
from gem5.components.boards.simple_board import SimpleBoard
from gem5.components.cachehierarchies.classic.private_l1_cache_hierarchy import PrivateL1CacheHierarchy
from gem5.components.memory.single_channel import SingleChannelDDR3_1600
from gem5.components.processors.cpu_types import CPUTypes
from gem5.components.processors.simple_processor import SimpleProcessor
from gem5.isas import ISA
from gem5.resources.resource import obtain_resource
from gem5.simulate.simulator import Simulator


# A simple script to test with different CPU models
# We will run a simple application (matrix-multiply) with AtomicSimpleCPU and TimingSimpleCPU
# using two different cache sizes
...
```

Let's start with Atomic CPU

`cpu_type` in cores.py should already be set to Atomic.

```
# By default, use Atomic CPU
cpu_type = CPUTypes.ATOMIC

# Uncomment for steps 2 and 3
# cpu_type = CPUTypes.TIMING
```

Let's run it!

```
cd /workspaces/2024/materials/02-Using-gem5/04-cores
gem5 --outdir=atomic-normal-cache cores.py
```

Make sure the out directory is set to **atomic-normal-cache**.

Next, try Timing CPU

Change `cpu_type` in cores.py to Timing.

```
# By default, use Atomic CPU
# cpu_type = CPUTypes.ATOMIC

# Uncomment for steps 2 and 3
cpu_type = CPUTypes.TIMING
```

Let's run it!

```
gem5 --outdir=timing-normal-cache cores.py
```

Make sure the out directory is set to **timing-normal-cache**.

Now, try changing the Cache Size

Go to this line of code.

```
cache_hierarchy = PrivateL1CacheHierarchy(l1d_size="32KiB", l1i_size="32KiB")
```

Change `l1d_size` and `l1i_size` to 1KiB.

```
cache_hierarchy = PrivateL1CacheHierarchy(l1d_size="1KiB", l1i_size="1KiB")
```

Let's run it!

```
gem5 --outdir=timing-small-cache ./materials/02-Using-gem5/04-cores/cores.py
```

Make sure the out directory is set to **timing-small-cache**.

Now let's try a Small Cache with Atomic CPU

Set `cpu_type` in cores.py to Atomic.

```
# By default, use Atomic CPU
cpu_type = CPUTypes.ATOMIC

# Uncomment for steps 2 and 3
# cpu_type = CPUTypes.TIMING
```

Let's run it!

```
gem5 --outdir=atomic-small-cache cores.py
```

Make sure the out directory is set to **atomic-small-cache**.

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Statistics

Look at the Number of Operations

Run the following command.

```
grep -ri "simOps" *cache
```

Here are the expected results. (Note: Some text is removed for readability.)

```
atomic-normal-cache/stats.txt:simOps                                    33954560
atomic-small-cache/stats.txt:simOps                                     33954560
timing-normal-cache/stats.txt:simOps                                    33954560
timing-small-cache/stats.txt:simOps                                     33954560
```

> "Ops" may be different from "Instructions" because gem5 breaks instructions down into "micro-ops."
> x86 is highly microcoded, all ISAs have some microcoded instructions in gem5.

Look at the Number of Execution Cycles

Run the following command.

```
grep -ri "cores0.*numCycles" *cache
```

Here are the expected results. (Note: Some text is removed for readability.)

```
atomic-normal-cache/stats.txt:board.processor.cores0.core.numCycles        38157549
atomic-small-cache/stats.txt:board.processor.cores0.core.numCycles         38157549
timing-normal-cache/stats.txt:board.processor.cores0.core.numCycles        62838389
timing-small-cache/stats.txt:board.processor.cores0.core.numCycles         96494522
```

Note that for Atomic CPU, the number of cycles is the **same** for a large cache *and* a small cache.

This is because Atomic CPU ignores memory access latency.

Extra Notes about gem5 Statistics

When you specify the out-directory for the stats file (when you use the flag `--outdir=<outdir-name>`), go to **`<outdir-name>/stats.txt`** to look at the entire statistics file.

For example, to look at the statistics file for the Atomic CPU with a small cache, go to **`atomic-small-cache/stats.txt`**.

In general, if you don't specify the out-directory, it will be **`m5out/stats.txt`**.

Other statistics to look at

- Simulated time (time simulated by gem5)
  - `simSeconds`
- Host time (time taken by gem5 to run your simulation)
  - `hostSeconds`

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Let's configure a custom processor!

Material to use

materials/02-Using-gem5/04-cores/cores-complex.py

materials/02-Using-gem5/04-cores/components/processors.py

Steps

1. Update class Big(O3CPU) and Little(O3CPU)
2. Run with Big processor
3. Run with Little processor
4. Compare statistics

We will be running the same workload (matrix-multiply) on **two custom processors**.

Configuring two processors

We will make one fast processor (***Big***) and one slow processor (***Little***).

To do this, we will change **4** parameters in each processor.

- **width**
  - Width of fetch, decode, rename, issue, wb, and commit stages
- **rob_size**
  - The number of entries in the reorder buffer
- **num_int_regs**
  - The number of physical integer registers
- **num_fp_regs**
  - The number of physical vector/floating point registers

Configuring Big

Open the following file: materials/02-Using-gem5/04-cores/components/processors.py

On the right, you'll see what `class Big` currently looks like.

Change the parameter values so that they are as follows:

- `width=10`
- `rob_size=40`
- `num_int_regs=50`
- `num_fp_regs=50`

```
class Big(O3CPU):
    def __init__(self):
        super().__init__(
            width=0,
            rob_size=0,
            num_int_regs=0,
            num_fp_regs=0,
        )
```

Configuring Little

Now, on the right, you'll see what `class Little` currently looks like.

Change the parameter values so that they are as follows:

- `width=2`
- `rob_size=30`
- `num_int_regs=40`
- `num_fp_regs=40`

```
class Little(O3CPU):
    def __init__(self):
        super().__init__(
            width=0,
            rob_size=0,
            num_int_regs=0,
            num_fp_regs=0,
        )
```

Run with Big processor

We will be running the following file.
materials/02-Using-gem5/04-cores/cores-complex.py

First, we will run matrix-multiply with our `Big` processor.

Run with the following command:

```
gem5 --outdir=big-proc cores-complex.py -p big
```

Make sure the out directory is set to **`big-proc`**.

Run with Little processor

Next, we will run matrix-multiply with our `Little` processor.

Run with the following command:

```
gem5 --outdir=little-proc cores-complex.py -p little
```

Make sure the out directory is set to **`little-proc`**.

Comparing Big and Little processors

Run the following command.

```
grep -ri "simSeconds" *proc && grep -ri "numCycles" *proc
```

Here are the expected results. (Note: Some text is removed for readability.)

```
big-proc/stats.txt:simSeconds                                    0.028124
little-proc/stats.txt:simSeconds                                 0.036715
big-proc/stats.txt:board.processor.cores.core.numCycles          56247195
little-proc/stats.txt:board.processor.cores.core.numCycles       73430220
```

Our `Little` processor takes more time and more cycles than our `Big` processor.